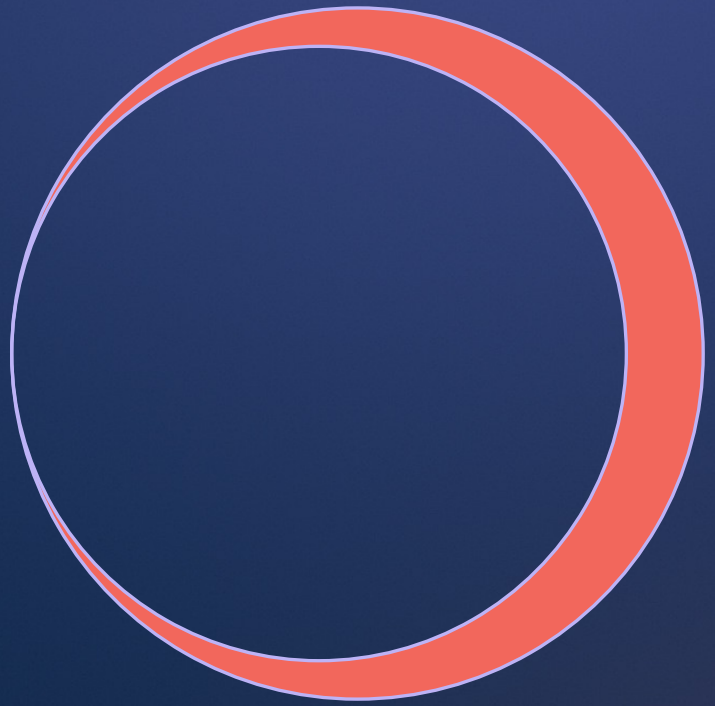
AIM HEALTHCARE HACKATHON VIENNA 2026 · TRACK 3 · EUROPEAN HEALTH DATA SPACE

Can two safe answers *reveal a secret?*



HealthCare Rocket 🚀

As fast as a rocket. Without the complexity.

THE RULE Counts below 10 are shown as “<10”

REQUEST A

Women aged 45 to 74 with breast cancer

AUSTRIA

72 ✓ Released

−

REQUEST B

Women aged 45 to 74 with breast cancer

AUSTRIA WITHOUT BURGENLAND

71 ✓ Released

=

1

One woman in Burgenland.

A count the rule would show as “<10”.

Both answers pass the rule.

Together, they break the protection.

Counts verified on the official synthetic hackathon dataset (20,000 patients) · ICD-10 C50 · age on 31 Dec 2025 · opted-out patients excluded

Allowed is not the same as *safe.*

Every request passes two checks before anything is released.

INPUT

Research question

in plain language

CHECK 1 — WORKING

Is the request allowed?

- Purpose permitted
- Statistics or secure access
- Minimum necessary data
- Opt-outs excluded

CHECK 2 — WORKING

Is the answer safe to release?

- Small cells hidden

A ledger records every release

72 released → 71 withheld

WORKING verified in tests

Answer

statistics or secure access

Ask

when information is missing

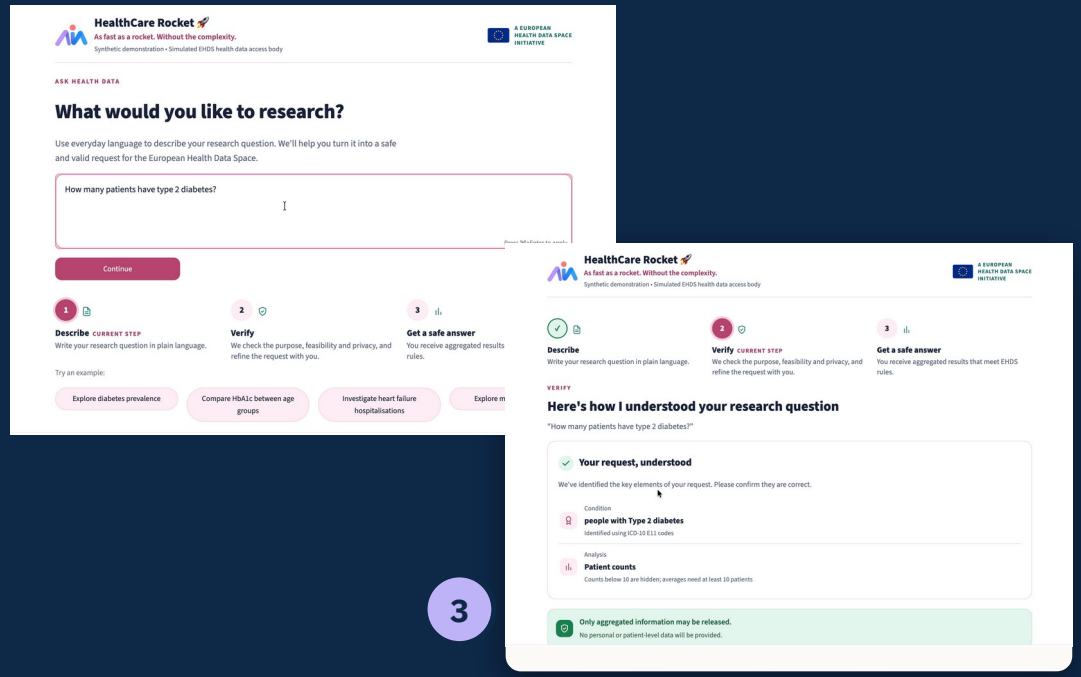
Say no

prohibited purpose or unsafe output

The complexity should be in the system, *not in the researcher's experience.*

Researchers should describe their question, not assemble JSON.

1. **Describe it in your own words** — WORKING
   No code, no JSON, no database terms
2. **See how it was understood** — WORKING
   A plain-language summary before it runs
3. **Fill in what is missing** — WORKING
   It asks for the purpose or a missing criterion
4. **Or build it step by step** — WORKING
   Question, population, analysis, review
5. **Receive authorised statistics** — WORKING
   Small counts hidden, articles cited



CURRENT PROTOTYPE SCREENSHOTS · SYNTHETIC DATA ONLY

TRUST BY DESIGN

Read the words. *Protect the data.*

INTERPRETATION — WORKING

Rule-based today: patterns and medical dictionaries

- ✓ Reads only the question
- × Never sees a patient record
- × Never calculates a number

OPTIONAL AI ADAPTERS — FUTURE

Mistral and Ollama adapters are in the code, not yet connected or tested with real models.

01 **Research question** — In the researcher's own words — WORKING

02 **Interpretation** — Rule-based: patterns and medical dictionaries — WORKING

03 **Structured query** — Shown back in plain language before it runs — WORKING

ONLY THE STRUCTURED QUERY CROSSES ↓

04 **Authorisation** — EHDS rules: Art. 53, 54, 66, 68, 69, 73 — WORKING

05 **Computation** — Deterministic statistics, identical on every run — WORKING

06 **Privacy enforcement** — Small cells, opt-outs, disclosure ledger — WORKING

07 **Result with reasons** — Decision and cited articles for applicant and staff — WORKING

CONTROLLED ENVIRONMENT · PATIENT-LEVEL DATA STAYS INSIDE

THE VISION

Making health data accessible *without making patients identifiable.*

FOR RESEARCHERS

Questions in their own words, answers they can trust.

FOR ACCESS BODIES

Consistent decisions, explained in plain language.

FOR PATIENTS

Protection that looks across answers, not only at one.

A gateway that knows when to answer, when to ask, and when to say no.

How the gateway *is built*

WORKING · PLANNED · FUTURE

APPLICANT SIDE

What researchers and staff see

- Plain-language entry
- Guided request builder
- Advanced options, JSON
- Applicant and purpose form
- Plain-language summary
- Decision with reasons

INTERPRETATION

Question text only, no patient data

- Rule-based interpreter
- Medical dictionaries
- Clarifying questions
- AI model adapters (future, untested)

CONTROLLED ENVIRONMENT

Patient-level data stays here

- Decision rules, Art. 53 to 73
- Deterministic statistics
- Small cells, opt-outs excluded
- Disclosure ledger
- SPE, download block (simulated)
- Audit record

20,000 PATIENTS · 839 OPTED OUT

Clinical words in, *standard codes out*

FROM THE OFFICIAL REQUESTS

Type 2 diabetes	ICD-10 E11
Atrial fibrillation	ICD-10 I48
Breast cancer	ICD-10 C50
Oral anticoagulant	ATC B01AA, B01AF
SGLT2 inhibitor	ATC A10BK
HbA1c	LOINC 4548-4
BMI	LOINC 39156-5

Codes match on the prefixes given in each request's definitions.

SAME WORDS, TWO COHORTS

"Hospitalised for heart failure in 2025"

103 patients whose stay had heart failure as the main diagnosis

123 patients with heart failure who had any inpatient stay

The gateway asks which one you mean.

WORKING

Official synthetic dataset · stays starting in 2025 · opt-outs excluded

CLINICAL REALISM

Synthetic data is cleaner than hospital data. Real records add free text, local codes, mixed units and missing context.

Before deployment

FUTURE

Harmonisation to OMOP or FHIR, unit checks, and validation on real-world data.

A working prototype is the beginning of clinical validation, not the end.

What we enforce, *and where it ends*

PER ANSWER — WORKING

- Counts below 10, including 0, shown as “<10”
- Percentages hidden if numerator or denominator is below 10
- Means hidden below 10 patients
- Every requested group shown, even if empty
- All 839 opted-out patients excluded

ACROSS ANSWERS — WORKING

- Built: a disclosure ledger records every released result
- Each new result is compared with everything released before, to any applicant
- Withheld when one population contains the other and they differ by fewer than 10, including 0
- Tested: subset and complement pairs, other applicants, reruns (pytest)

WHERE IT ENDS

- Overlapping, non-nested queries
- Chains of three or more answers
- Several applicants combining results
- Outside knowledge about individuals

NEXT — FUTURE

Query budgets, staff output checks, noise for high-risk outputs

VERIFIED ON THE OFFICIAL SYNTHETIC DATASET

Women aged 45 to 74 with breast cancer: 72 − 71 = 1 · All ages: 114 − 112 = 2 · Official R08 answer: Burgenland “<10”

Measured results, *then the road ahead*

OFFICIAL CHALLENGE METRICS · THE 10 REQUESTS

Correct decisions
Target 10 of 10
Ours: 10 of 10

Correct numbers
Target 6 of 6
Ours: 6 of 6

Privacy leaks
Target 0
Ours: 0

Consistency
Target 10 of 10
Ours: 10 of 10

Slowest answer
Target under 60 s
Ours: 0.25 s

Our own tests
208 automated tests pass: new requests, reworded purposes, prompt injection, differencing pairs, UI flows

ROADMAP

01 **Today**
Working end to end: plain-language entry, interpretation, EHDS decisions, statistics, disclosure ledger. HDAB and SPE simulated.

02 **Next** PLANNED
Harden the ledger: chains of three or more answers, means, cumulative release accounting

03 **Then** FUTURE
Test the optional AI adapters against the rule-based interpreter, with documented data flows

04 **Pilot** FUTURE
With access-body staff and researchers, on harmonised real-world data